

“X’y O’sies” the Game

The Irish have a game called “X’y O’sies” (in Gallic) or **Xs And Os** for the rest of us. The English call it **“noughts and crosses”**, but you might know it by another name. It is a game for two players, X and O, who take turns marking the spaces in a 3×3 grid. The player who succeeds in placing three of their marks in a diagonal, horizontal, or vertical row is the winner. The Specification for the code is given bellow and some code for the game is started. You need to look over the code and add what is needed.

Files

- XsAndOs.java - starter code (add your code to this one)
- Game.java - play a game of Xs And Os.
- TheTester.java - a brutal tester to make sure your isWinner() method is **always** correct

XsAndOs
- borad : int [][] - turns : int
+ XsAndOs() + isWinner(p: char) : boolean + isFull(): boolean + isCat() : boolean + isValid(r:int, c:int): boolean + numTurns(): int + playerAt(r:int, c:int) : char + displayBoard(): void + playMove(p : char, r: int, c : int) : void

Instance Variables	
board	two dimensional array of chars
turns	integer to keep track of the number of turns played so far
Constructors	
XsAndOs()	default constructor: 1) creates the two-dimensional array 2) fills each slot with ' ' (a blank space) 3) initializes the other attributes
Accessors	
boolean isWinner(char p)	returns <code>true</code> if the given letter currently has three in a row and <code>false</code> otherwise. That is, <code>isWinner('X')</code> will return true if X has won
boolean isFull()	returns <code>true</code> if 9 turns have been played and <code>false</code> otherwise
boolean isCat()	returns <code>true</code> if all 9 spaces of the grid are taken AND neither X nor O has won, <code>false</code> otherwise. (a cat game tie or draw)
boolean isValid(int r, int c)	returns <code>true</code> if the given row and column corresponds to a valid space on the board
int numTurns()	returns the <code>number</code> of turns played so far
char playerAt(int r, int c)	returns the <code>character</code> representing the piece at the given location. Should return either ' ', 'X', or 'O'.
void displayBoard()	displays the current board on the screen
Modifiers	
void playMove(char p, int r, int c)	allows the given player to place their move at the given row and column. The row and column numbers are 0-based, so valid numbers are 0, 1, or 2

```
[malcolm@ solution]$ javac XsAndOs.java Tester.java && java Tester
Checking constructor..... [ok]
Checking isValid()..... [ok]
Checking isWinner()..... [ok]

All tests passed!

[malcolm@ solution]$
```

```
[malcolm@ solution]$ javac Game.java
[malcolm@ solution]$ java Game
 0  | |
  --+--
 1  | |
  --+--
 2  | |
    0 1 2
'X', choose your location (row, column): 1 1
 0  | |
  --+--
 1  |X|
  --+--
 2  | |
    0 1 2
'O', choose your location (row, column): 0 2
 0  | |O
  --+--
 1  |X|
  --+--
 2  | |
    0 1 2

'X', choose your location (row, column): 0 0
 0  X| |O
  --+--
 1  |X|
  --+--
 2  | |
    0 1 2
'O', choose your location (row, column): 1 2
 0  X| |O
  --+--
 1  |X|O
  --+--
 2  | |
    0 1 2
'X', choose your location (row, column): 2 2
 0  X| |O
  --+--
 1  |X|O
  --+--
 2  | |X
    0 1 2
X is the winner!
```